Points of view

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The companion website

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Check for updates

How biologists can make the most of supplementary materials for impact.

"Overview first, details on demand" is a well-known mantra in data visualization[1]. While this is argued for visual depictions of data, we find an analogy for papers as well. An abstract gives a quick overview of the essence of a paper; some journals require graphical abstracts[2] or may allow just a few essential figures in the paper that present its main evidence. Further details are often placed in supplementary figures. Today, however, websites can also serve as supplementary material, with relatively little effort.

From static supplements to dynamic research artifacts

When writing papers, authors go through many iterations of figures, tables and other display items. Only the most essential make it into the main body of the manuscript, so the final paper typically represents only a small fraction of the outputs generated during a long preprocessing, analysis and visualization process. Supplementary materials have long been an established way to document results that support the manuscript, provide more details or present unaltered views of scientific data, such as immunoblots. They typically include tables and figures, alongside further references or methodological details.

In most cases, journals host these figures in static PDFs. This ensures long-term availability and wide usability, but this static format also limits researchers to figures, tables and text that cannot easily convey dynamic or interactive information. Many studies produce animated visualizations, interactive plots or three-dimensional (3D) models that do not appear in papers, not because they lack quality or relevance, but because journals cannot easily publish them.

Occasionally, journals encourage the deposition of code capsules (such as on Code Ocean) or the use of dedicated data repositories such as FigShare or Zenodo. These resources focus primarily on the unaltered data and code necessary for reproducibility – for example, sequencing reads or microscopy images – rather than on figures designed to help readers explore and understand the results.

Companion websites for dynamic supplementary figures

To overcome the limitations of traditional supplementary items, companion websites can be created with relatively little effort today. These can function as dynamic, rich repositories for additional content, especially figures. They enable the authors to embed interactive visualizations, videos and annotated figures that are impractical to publish in static PDF form. They can also help readers navigate complex datasets and analyses by providing exploratory interfaces. Researchers may already be producing figures suitable for such sites during the research process; interactive plots, animated visualizations or exploratory tools can be valuable when shared alongside the paper. Figure 1 illustrates different interactive visuals that can bring supplementary materials to life on a companion website.

Tools

Creating a companion website does not necessarily require extensive web development expertise. In many cases, a simple framework providing layout and structure is sufficient, allowing text to be combined with images, videos or interactive plots. Markdown is a lightweight markup language for documents that can be learned in minutes. Self-hosted Markdown environments, such as HackMD, already support many use cases. Markdown documents can also embed HTML elements, which enables the inclusion of YouTube videos, images and 3D viewers. Using Iframes, almost any HTML content can be embedded, including interactive data visualizations.

Web components further expand these possibilities. Many web-based applications are now available as lightweight components that can be reused across the web and integrated into companion websites. For example, Human Reference Atlas[3] user interfaces are available as web components, which enable spatial and semantic exploration of data registered within the 81 three-dimensional reference organs of the HRA v2.5. Code for embedding these web interfaces is available.

Lightweight deployment with GitHub Pages

One accessible way to deploy companion websites is provided by GitHub Pages. GitHub offers native support for generating HTML websites directly from Markdown documents, which allows scientists to write content in a workflow similar to rich-text editing while maintaining lightweight, interoperable files and version control on a platform many already use.

Several recent publications illustrate how such approaches can enhance supplementary figures:

- An integrated 3D model viewer component alongside an embedded Sankey diagram illustrate which data were and were not included in the analysis for the manuscript[4]
- A companion website uses an embedded video tutorial, annotated screenshots on using an API, tables linking to database queries, and Scalable Vector Graphics (SVG) diagrams generated from Mermaid.js scripts to explain the data structures underlying a biological knowledge graph[5]
- A third example presents Jupyter notebooks using Quarto to support a meta-study on cell-to-cell distances in spatial omics datasets[6], which allows readers to explore the analysis interactively
- Content-aware image restoration (CARE)[7] uses a companion website to showcase a plethora of images with metadata on authorship, details, demos, and videos
- The team behind the Dresden Ovary Table (DOT)[8] uses a companion website to host data on RNA localization for more than 1,700 genes and provides derived 3D microscopy of the adult fruit fly ovary over time

Practical considerations

Companion websites should complement, not duplicate, the figures in the manuscript. Repeating large sections of text from the paper should be avoided; the site should remain concise and focused on figures that benefit from interactive presentation.

File sizes, especially for embedded figures, should remain manageable so the companion website remains accessible for readers with low internet bandwidth. Only the most relevant figures should be included. All figures

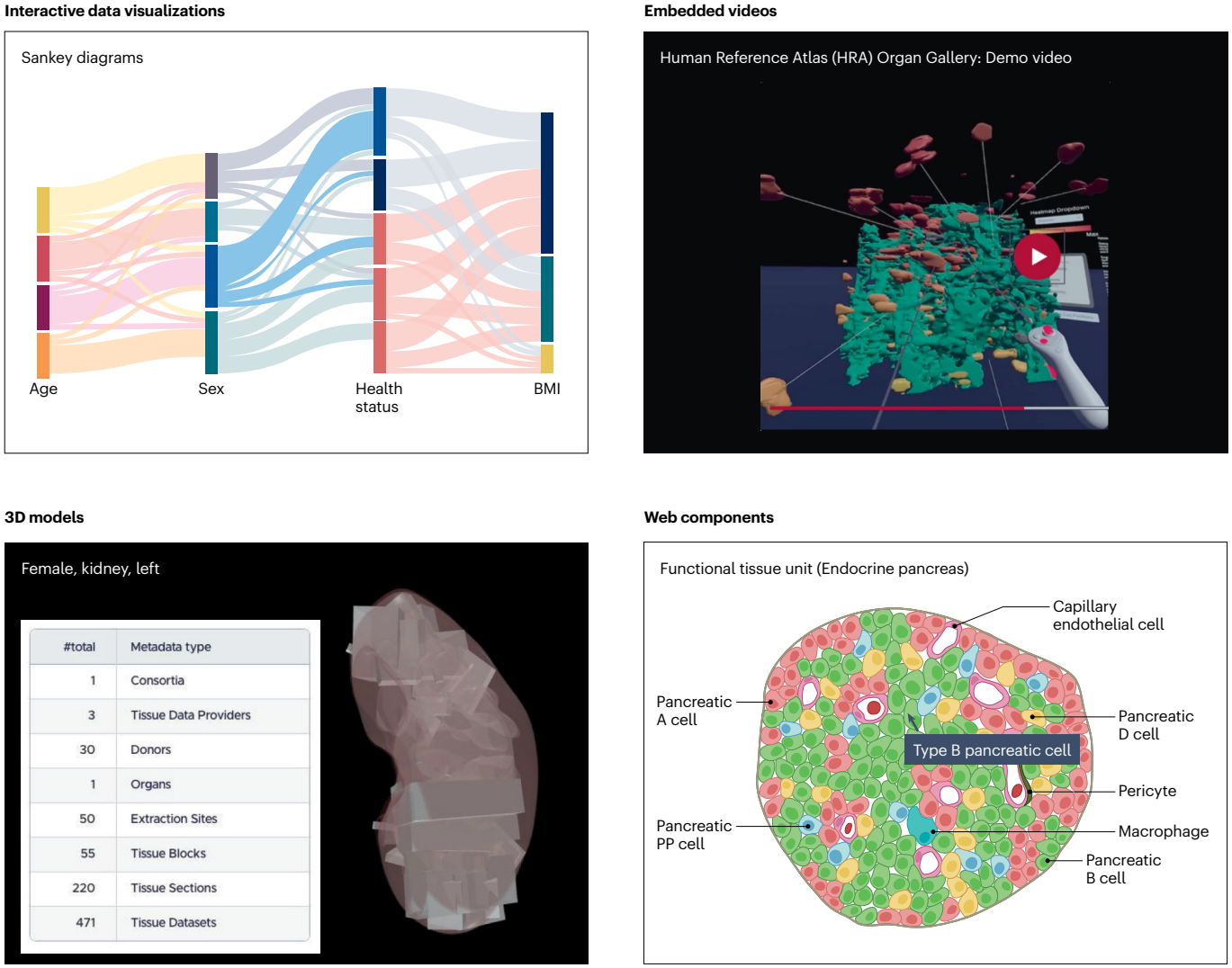


Fig. 1 | An illustration of interactive content types on a companion website. Top left: an embedded interactive Sankey diagram. Top right: an embedded video showcasing a virtual reality visualization with voice-over by the data provider[10]. Bottom left: an embedded 3D model[3] with an auxiliary data table. Bottom right: a web component showing an interactive 2D illustration[11] that reacts to user input (hovering).

and tables should meet publication-quality standards, even though they may not be formally assessed during peer review.

Longevity is another important consideration. Commercial platforms such as GitHub provide convenient hosting and long-term archival stability, especially when using free tiers. Researchers should also consider other strategies to ensure continued accessibility, such as hosting companion pages on servers provided by their own institutions. Keeping companion websites light is an important consideration for longevity.

To ensure adherence to FAIR (findable, accessible, interoperable and reusable) data principles[9], any post-publication updates should be made cautiously and carefully documented. An official change log on the website can work well, as does a clear indication of the last update in the footer date to maintain transparency. When using GitHub, the commit history provides additional transparency, but readers should not need to navigate the repository to understand changes.

Ideally, companion websites should be available during the review process — for example, alongside a preprint submission — and updated as the manuscript evolves before being frozen in time once the paper is published.

Expanding the reach of supplementary materials

Companion websites provide a practical way to share high-quality research data that are impractical for publication in traditional print, online or hybrid journals, extending their life and accessibility and transforming them from static appendices into interactive research resources. By building on widely used tools such as Markdown and GitHub, they can be created with minimal overhead. Once published, they can be linked from the paper itself, as well as from presentations, talks or other contexts where the research is discussed.

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Competing interests

The authors declare no competing interest.